
Online Food Delivery System

Software Engineering
Project (FR. Part)



Requirements Engineering:

Functional Requirements :

FR-01 – User Registration & Login:

User Requirement : A customer shall be able to create a new account by providing their full name, email address, and password, and shall be able to log in using their credentials to access the platform.

System Requirement : The system shall validate the email format and check it is not already registered. It shall hash the password before storing it. Upon successful login, the system shall generate a session token to maintain the authenticated state.

Priority : High

Actor : Customer

Rationale : This is the entry point for all system functionality. Without authentication, no personalized features can be accessed.

Pre-condition : The user has the app installed and is not currently logged in.

Post-condition : The user is authenticated and redirected to the home screen.

Normal Flow :

- 1. User opens the app and selects "Sign Up" or "Log In"**
- 2. User enters their details and submits the form**
- 3. System validates the input**
- 4. System creates the account or verifies credentials**
- 5. System grants access and loads the home screen**

Alternative Flow : If the email is already registered, the system displays an error and prompts the user to log in instead. If credentials are incorrect, the system allows the user to retry or reset their password.

FR-02 – Browse Restaurants

User Requirement : A customer shall be able to browse a list of available restaurants near their location and filter results by cuisine type, rating, or estimated delivery time.

System Requirement : The system shall retrieve the user's GPS coordinates and query the database for restaurants within a configurable delivery radius. It shall apply the selected filters and return a sorted list of matching restaurants.

Priority : High

Actor : Customer

Rationale : Browsing is the first step in the ordering process. Without it, customers cannot discover available options.

Pre-condition : The user is logged in and has location access enabled.

Post-condition: A filtered list of nearby restaurants is displayed to the user.

Normal Flow :

1. User opens the home screen
2. System detects user location automatically
3. System fetches and displays nearby restaurants
4. User applies filters if desired
5. System refreshes the list based on selected filters

Alternative Flow : If location access is denied, the system prompts the user to enter their address manually. If no restaurants match the filters, the system displays a "No results found" message and suggests removing filters.

FR-03 – View Restaurant Menu

User Requirement : A customer shall be able to select a restaurant and view its full menu, including item names, descriptions, prices, and availability.

System Requirement : The system shall fetch the menu data associated with the selected restaurant from the database and render it grouped by category. Items marked as unavailable by the restaurant shall be displayed as greyed out and non-selectable.

Priority : High

Actor : Customer

Rationale : Viewing the menu is essential before any order can be placed. Customers need complete and accurate item information to make purchasing decisions.

Pre-condition : The user has selected a restaurant from the browse screen.

Post-condition: The restaurant's menu is fully displayed with current availability status.

Normal Flow :

1. User taps on a restaurant from the list
2. System retrieves the menu from the database
3. System displays items grouped by category (e.g. Appetizers, Main Course, Drinks)
4. User browses items and views details

Alternative Flow : If the menu fails to load due to a network error, the system displays an error message and offers a retry option. If the restaurant has temporarily closed, the system notifies the user and disables the ordering option.

FR-04 – Manage Shopping Cart

User Requirement : A customer shall be able to add items to a cart, adjust quantities, remove items, and view the total price before placing an order.

System Requirement : The system shall maintain a session-based cart that stores item IDs, selected quantities, unit prices, and a calculated running total including any applicable fees or discounts. The cart shall persist across screens during the same session.

Priority : High
Actor : Customer

Rationale : The cart acts as the bridge between browsing and ordering. It allows customers to review and confirm their selection before committing to a purchase.

Pre-condition : The user has viewed a restaurant menu and selected at least one item.
Post-condition: The cart reflects the correct items, quantities, and total price.

Normal Flow :

1. User taps "Add to Cart" on a menu item
2. System adds the item and updates the running total
3. User adjusts quantities or removes items as needed
4. System recalculates the total after each change
5. User proceeds to checkout when satisfied

Alternative Flow : If the user attempts to add items from a different restaurant while the cart already contains items, the system warns the user and asks whether to clear the existing cart or cancel the action.

FR-05 – Place Order

User Requirement : A customer shall be able to place an order from their cart and receive an order confirmation along with an estimated delivery time.

System Requirement : The system shall validate that all cart items are still available, create an order record in the database with a unique order ID, calculate the estimated delivery time, and send a confirmation notification to the customer. It shall also notify the restaurant of the new order.

Priority : High
Actor : Customer, System, Restaurant

Rationale : Placing an order is the core transaction of the entire system. All other features support or follow from this action.

Pre-condition : The user has a non-empty cart and has completed the payment step.
Post-condition : An order record is created, the restaurant is notified, and the customer receives a confirmation.

Normal Flow :

1. User reviews cart and taps "Place Order"
2. System validates item availability
3. System processes the payment
4. System creates the order record and assigns a unique ID
5. System sends confirmation to the customer with estimated delivery time
6. System notifies the restaurant

Alternative Flow : If an item becomes unavailable between cart addition and order placement, the system alerts the user and removes the item from the cart before proceeding. If payment fails, the order is not created and the user is prompted to try again.

FR-06 – Real-Time Order Tracking

User Requirement : A customer shall be able to track the status of their order in real time after it has been placed, from preparation through to delivery.

System Requirement : The system shall maintain and update an order status field that progresses through defined states: Confirmed, Preparing, Ready for Pickup, Out for Delivery, and Delivered. Status changes shall be pushed to the customer's interface without requiring a manual refresh.

Priority : High

Actor : Customer, System, Driver

Rationale : Real-time tracking increases customer trust and reduces the need for support inquiries about order status.

Pre-condition : The customer has successfully placed an order.

Post-condition : The customer can see the current status of their order updated in real time.

Normal Flow :

1. Customer opens the order tracking screen
2. System displays the current order status
3. Restaurant updates status to "Preparing"
4. Driver updates status to "Out for Delivery"
5. System marks the order as "Delivered" upon completion

Alternative Flow : If the driver is delayed, the system updates the estimated delivery time accordingly and notifies the customer. If the connection is lost, the system displays the last known status and attempts to reconnect automatically.

FR-07 – Payment Processing

User Requirement : A customer shall be able to pay for their order using a credit card, debit card, or digital wallet, and shall receive a payment confirmation upon success.

System Requirement : The system shall integrate with a third-party payment gateway to securely process transactions. It shall not store raw card details and shall use tokenization for payment data. A payment receipt shall be generated and sent to the user's registered email upon successful transaction.

Priority : High

Actor : Customer, System

Rationale : Payment processing is a critical step in completing any order. The system must handle it securely and reliably to protect user financial data.

Pre-condition : The user has reviewed their cart and is ready to confirm the order.

Post-condition : The payment is processed, a receipt is issued, and the order proceeds to the restaurant.

Normal Flow :

1. User selects a payment method at checkout
2. User enters payment details or selects a saved method
3. System sends the transaction to the payment gateway
4. Payment gateway returns a success response
5. System confirms the order and sends a receipt to the user

Alternative Flow : If the payment gateway returns a failure response, the system notifies the user with the reason and allows them to retry with the same or a different payment method. If a timeout occurs, the system informs the user and does not create an order record.

FR-08 – Restaurant Dashboard & Menu Management

User Requirement : A restaurant owner shall be able to log in to a dedicated dashboard where they can add, edit, or remove menu items and mark items as available or unavailable.

System Requirement : The system shall provide a restaurant-facing interface with role-based access control. Changes made to the menu shall be written to the database immediately and reflected in the customer-facing app in real time.

Priority : Medium

Actor : Restaurant Owner

Rationale : Restaurants need to maintain accurate and up-to-date menus. Without this feature, customers may order unavailable items, leading to cancellations and poor experience.

Pre-condition : The restaurant owner has a verified account with restaurant-level permissions.

Post-condition : Menu changes are saved and immediately visible to customers browsing the restaurant.

Normal Flow :

1. Restaurant owner logs in to the dashboard
2. Owner navigates to the menu management section
3. Owner adds a new item, edits an existing one, or marks one as unavailable
4. System saves the changes to the database
5. Updated menu is immediately visible in the customer app

Alternative Flow : If required fields such as item name or price are missing, the system displays a validation error and prevents saving. If the session expires, the owner is redirected to the login page without losing unsaved changes.

FR-09 – Delivery Driver Assignment

User Requirement : Once an order is ready for pickup, the system shall automatically assign an available delivery driver to collect and deliver the order.

System Requirement : The system shall query the location data of all currently active and unassigned drivers. It shall calculate proximity to the restaurant and assign the nearest available driver. The driver shall receive a push notification with the order and pickup details.

Priority : High

Actor : System, Driver

Rationale : Automated driver assignment reduces manual coordination overhead and ensures timely delivery by selecting the most geographically suitable driver.

Pre-condition : The restaurant has confirmed the order is ready for pickup and at least one driver is available.

Post-condition: A driver is assigned to the order and notified with full pickup details.

Normal Flow :

1. Restaurant marks the order as "Ready for Pickup"
2. System queries all available drivers and their current locations
3. System selects the nearest driver
4. System sends a push notification to the assigned driver
5. Driver accepts the assignment and proceeds to the restaurant

Alternative Flow : If no drivers are available, the system retries every 60 seconds and notifies the customer of the delay. If the assigned driver rejects the order, the system reassigns to the next nearest available driver.

FR-10 – Ratings & Reviews

User Requirement : A customer shall be able to rate a restaurant on a scale of 1 to 5 stars and optionally write a text review after an order has been delivered.

System Requirement : The system shall allow exactly one review submission per completed order. Upon submission, it shall store the rating and review text, update the restaurant's overall average rating, and display the review on the restaurant's public profile.

Priority : Medium

Actor : Customer

Rationale : Ratings and reviews help customers make informed choices and provide restaurants with feedback to improve their service quality.

Pre-condition : The order status has been marked as "Delivered."

Post-condition : The review is saved, the restaurant's average rating is updated, and the review appears on the restaurant's profile.

Normal Flow :

1. Customer receives a prompt to rate the order after delivery
2. Customer selects a star rating and optionally writes a review
3. Customer submits the review
4. System stores the review and recalculates the restaurant's average rating
5. Review is published on the restaurant's profile page

Alternative Flow : If the customer dismisses the prompt, they can still access the review option from their order history. If the customer attempts to submit a review for the same order twice, the system rejects the second submission and displays a message indicating a review already exists.

FR-II – Order History & Reorder

User Requirement : A customer shall be able to view a complete history of their past orders and quickly reorder items from a previous purchase with a single action.

System Requirement : The system shall retrieve all completed order records linked to the authenticated user's account, sorted by date in descending order. When the reorder action is triggered, the system shall populate the cart with the same items from the selected past order, subject to current availability.

Priority : Medium

Actor : Customer

Rationale : Order history improves convenience and encourages repeat usage by reducing the effort required to place recurring orders.

Pre-condition : The user is logged in and has at least one completed order.

Post-condition : The selected past order's items are added to the cart, ready for checkout.

Normal Flow :

1. User navigates to the "Order History" section
2. System retrieves and displays all past orders sorted by date
3. User selects a past order to view its details
4. User taps "Reorder"
5. System adds available items to the cart and notifies the user if any items are no longer available

Alternative Flow : If all items from the past order are unavailable, the system informs the user and does not populate the cart. If the restaurant from the past order is no longer active on the platform, the system displays a message and suggests similar alternatives.